

NAIPUNNYA SCHOOL OF MANAGEMENT

Cherthala, Alappuzha, Kerala
Affiliated to University of Kerala

**SUMMER INTERNSHIP REPORT
ON**

AI WITH CYBERSECURITY

AT

LOGICCRAFT SOFTWARE SOLUTIONS

Report Submitted to the University of Kerala in partial fulfillment of the
requirements for the 4th Semester FYUGP BSc Computer Science

Submitted by

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NAIPUNNYA SCHOOL OF MANAGEMENT

Cherthala, Alappuzha – 688 524

2026

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BONAFIDE CERTIFICATE

This is to certify that this internship report entitled "AI with Cybersecurity" is the bonafide work carried out by Alan Thomas, Register No. 4537, APAAR ID: 216786261499, student of 4th Semester BSc Computer Science, Naipunnya School of Management, Cherthala, during the academic year 2025–2026.

The internship was undertaken at Logiccraft Software Solutions for a duration of 26 Days during the academic year 2025–2026. The report is submitted to the University of Kerala in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Computer Science.

Ms. Gayathri A.

Faculty Guide

Head of Department

Principal

CERTIFICATE FROM INTERNSHIP ORGANISATION

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Alan Thomas (BSc Computer Science), Dept. of Computer Science & Applications, Naipunnya School of Management, Cherthala, has successfully completed a 26-Day Internship in AI with Cybersecurity with our organisation during the academic year 2025–2026.

During the internship period, the intern successfully completed tasks related to the following areas:

- Cybersecurity Fundamentals and the CIA Triad
- Network Reconnaissance and Packet Analysis using Wireshark
- Python Programming for Security Automation and Log Monitoring
- Keylogger Awareness Project using Python, Regex, and SQLite
- Machine Learning-based Intrusion Detection System (IDS) using CNN and LSTM

We found the intern to be diligent, sincere, and committed to the responsibilities assigned. We wish the intern continued success in all future endeavors.

For Logiccraft Software Solutions

Authorized Signatory

Logiccraft Software Solutions

DECLARATION

I, Alan Thomas, hereby declare that the Summer Internship Report titled "AI with Cybersecurity" carried out at Logiccraft Software Solutions is a genuine record of the work done by me during the academic year 2025–2026 in partial fulfillment of the requirements for the 4th Semester FYUGP BSc Computer Science programme under the Department of Computer Science & Applications, Naipunnya School of Management, Cherthala. I further declare that this report has not been submitted previously, either fully or partially, for the award of any degree, diploma, or other similar title in any institution.

Alan Thomas

S4 FYUGP BSc Computer Science

Naipunnya School of Management, Cherthala

Forwarded to the University

Faculty Mentor – Name and Signature

IPO Supervisor – Name and Signature

Head of the Department – Name and Signature

Principal – Name and Signature

Place: Cherthala

Date: _____

Seal

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who contributed to the successful completion of this internship and the preparation of this report.

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I extend my heartfelt thanks to the Head of the Department of Computer Science & Applications for their constant administrative facilitation and academic encouragement throughout the internship period.

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I am immensely thankful to Logiccraft Software Solutions for providing a structured, comprehensive, and industry-oriented internship programme. The mentors and technical staff at Logiccraft demonstrated extraordinary expertise and patience in guiding me through complex domains including cybersecurity fundamentals, Python security automation, and AI-driven intrusion detection.

Finally, I am deeply grateful to my family and friends for their unconditional support and motivation throughout all stages of my academic pursuits.

Alan Thomas

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AUC	Area Under the Curve
CIA	Confidentiality, Integrity, and Availability
CIDR	Classless Inter-Domain Routing
CNN	Convolutional Neural Network
DDoS	Distributed Denial of Service
DNS	Domain Name System
HTTP/S	Hypertext Transfer Protocol / Secure
IDS	Intrusion Detection System
IoC	Indicator of Compromise
LIME	Local Interpretable Model-Agnostic Explanations
LSTM	Long Short-Term Memory
ML	Machine Learning
MITRE ATT&CK	MITRE Adversarial Tactics, Techniques & Common Knowledge
NSL-KDD	Network Security Lab – Knowledge Discovery & Data Mining
OSI	Open Systems Interconnection
OTP	One-Time Password
OWASP	Open Web Application Security Project
Regex	Regular Expression
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SIEM	Security Information and Event Management
SMOTE	Synthetic Minority Over-sampling Technique
SOC	Security Operations Center
SQL	Structured Query Language
SSH	Secure Shell
TCP	Transmission Control Protocol
TLS	Transport Layer Security

UNSW-NB15	University of New South Wales Network Benchmark 2015
XSS	Cross-Site Scripting

CHAPTER 1 – INTRODUCTION

1.1 Background and Purpose of Internship

The twenty-first century has witnessed an unprecedented acceleration in digital transformation across every sector of human activity. While this revolution has unlocked extraordinary opportunities for efficiency and connectivity, it has simultaneously created an ever-expanding landscape of vulnerabilities. Cyber threats have grown in sophistication, volume, and economic impact, with nation-state actors, organized criminal groups, and independent attackers exploiting weaknesses in systems that billions of people and organizations depend on daily.

In this context, the convergence of Artificial Intelligence (AI) and Cybersecurity has emerged as one of the most critical frontiers in modern computer science. Traditional, rule-based security mechanisms are increasingly inadequate against polymorphic malware, zero-day exploits, and adaptive adversarial techniques. AI-driven approaches can learn from data, identify subtle behavioral patterns, and adapt to novel threats in ways that static systems cannot. This internship at Logiccraft Software Solutions was designed to provide hands-on exposure to precisely this intersection — immersing students in real-world toolsets, industry-standard frameworks, and practical project development across a structured 26-day programme.

1.2 Objectives of the Internship

The internship was designed with a clear and structured set of learning objectives aligned with both the academic requirements of the FYUGP BSc Computer Science programme and the practical demands of the cybersecurity industry:

- To master the fundamental principles of information security — specifically the CIA Triad (Confidentiality, Integrity, Availability) — and their application in risk assessment and secure system design.
- To gain technical proficiency in Python programming for security automation, focusing on defensive scripting, file integrity verification, log monitoring, and SSH administration.

- To explore and understand the OWASP Top 10 web application security risks, including SQL Injection, XSS, and insecure deserialization, along with their mitigation strategies.
- To develop practical skills in network reconnaissance and packet analysis using Wireshark, Nslookup, Dig, and Traceroute for anomaly detection and infrastructure mapping.
- To implement CNN and LSTM deep learning models for proactive network intrusion detection trained on the NSL-KDD and UNSW-NB15 benchmark datasets.
- To design and develop a Keylogger Awareness system using Python and Regular Expressions, demonstrating the risks of unencrypted endpoint inputs.
- To apply the MITRE ATT&CK framework for mapping adversary tactics and techniques to defensive countermeasures, and to develop professional technical documentation skills.

1.3 Scope of Work

The scope of the internship was deliberately broad yet cohesive, centered on the convergence of Artificial Intelligence and Cybersecurity across three progressive phases:

- Phase 1 – Cybersecurity and Networking Fundamentals (Days 1–16): Study of the CIA Triad, vulnerability assessment, OWASP Top 10, TCP/IP, OSI, DNS, TLS, network reconnaissance, Python security scripting, log monitoring, Wireshark analysis, and web security assessment.
- Phase 2 – Keylogger Awareness Project (Days 17–23): End-to-end development of a Python-based keystroke monitoring and sensitive data detection system using Regex and SQLite, culminating in a visualization dashboard.
- Phase 3 – AI-Based Intrusion Detection System (Days 24–26): Development, training, and evaluation of CNN and LSTM models on real security benchmark datasets, with SHAP/LIME explainability analysis.

1.4 Internship Details

Organization	Logiccraft Software Solutions
Duration	26 Days
Academic Year	2025–2026
Student Name	Alan Thomas

Register Number	4537
APAAR ID	216786261499
Course Code	UK4INTCSC200
Programme	4th Semester FYUGP BSc Computer Science
Faculty Guide	Ms. Gayathri A., Asst. Professor, Dept. of CS & Applications
Institution	Naipunnya School of Management, Cherthala

CHAPTER 2 – ORGANIZATION PROFILE

2.1 About Logiccraft Software Solutions

Logiccraft Software Solutions is a specialized, industry-oriented technology training and software development organization with a focused mandate: to prepare the next generation of technology professionals for the rapidly evolving challenges of modern computing. Operating at the intersection of academia and industry, Logiccraft has established itself as a trusted partner for students, educational institutions, and enterprises seeking to build practical competency in cutting-edge domains such as Artificial Intelligence, Machine Learning, and Cybersecurity.

The organization's flagship offering is its project-based internship programme, which is meticulously designed to simulate the conditions of a professional technology environment. Logiccraft's internship model emphasizes learning through doing — participants implement real tools, work with authentic datasets, and deliver functional projects that demonstrate measurable technical outcomes. Faculty and mentors are practitioners with extensive industry experience in Security Operations, Machine Learning Engineering, and Software Development, ensuring mentorship grounded in professional reality rather than purely theoretical instruction.

2.2 Vision and Mission

Vision

To cultivate a generation of technically proficient, ethically grounded, and professionally adaptable technology professionals capable of navigating the complexities of the global technology landscape with confidence, competence, and integrity.

Mission

To provide high-impact, practically immersive training in Artificial Intelligence and Cybersecurity, ensuring that academic knowledge is successfully translated into industry-ready skills through hands-on project work, expert mentorship, and real-world problem-solving. The organization is committed to designing curricula aligned with current industry demands, providing access to professional-grade tools and frameworks, fostering a culture of ethical responsibility, and building long-term partnerships with academic institutions.

2.3 Services and Offerings

- **Advanced Internship Programmes:** Structured, project-based internship programmes in AI-driven Cybersecurity, Data Science, Cloud Computing, and Software Development, designed to meet undergraduate and postgraduate credit requirements.
- **Technical Training Workshops:** Intensive short-duration workshops on emerging technologies including Deep Learning, Malware Analysis, SIEM implementation, and Penetration Testing, for students, faculty, and working professionals.
- **Software Development Services:** Custom software development and technical consulting services for local and regional clients, spanning web application development, database design, and process automation.
- **Academic Partnership Programmes:** Collaboration with colleges and universities to co-design internship curricula, provide guest lectures, and facilitate industry exposure through structured mentorship.

2.4 Why AI and Cybersecurity?

The decision to focus the curriculum on AI and Cybersecurity reflects three key strategic considerations. First, the escalating global cyber threat landscape — with cybercrime damages projected to reach trillions of dollars annually — creates urgent demand for skilled security professionals. Second, the fundamental limitations of traditional signature-based security tools against modern threats like polymorphic malware and Advanced Persistent Threats (APTs) make AI-driven detection not a luxury but a necessity. Third, professionals with combined expertise in AI and Cybersecurity are among the most sought-after globally, making this skill combination a strong career differentiator.

2.5 Technology Stack Used During the Internship

Category	Tools / Technologies
Programming Languages	Python 3.12 — primary language for automation, scripting, and ML development
ML / DL Libraries	Scikit-learn (Classical ML), TensorFlow / Keras (CNN, LSTM deep learning)
Explainability	SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations)
Security Frameworks	MITRE ATT&CK (adversary tactic and technique mapping)

Network Tools	Wireshark (packet capture), Snort (rule-based IDS), Nslookup / Dig / Traceroute
Log Management	Splunk Community Edition (log ingestion, correlation, dashboards)
Data & Storage	SQLite (event storage), Jupyter Notebook (EDA and model prototyping)
Datasets	NSL-KDD and UNSW-NB15 (benchmark network security datasets)

CHAPTER 3 – INTERNSHIP ACTIVITIES

3.1 Programme Overview

The internship at Logiccraft Software Solutions was structured as a comprehensive 26-day programme, progressing through three distinct phases. Phase 1 (Days 1–16) established a solid foundation in cybersecurity principles, networking, and Python-based security scripting. Phase 2 (Days 17–23) focused entirely on the Keylogger Awareness Project — a hands-on application of Python, Regular Expressions, and SQLite for endpoint security demonstration. Phase 3 (Days 24–26) involved the design, development, and evaluation of a Machine Learning-based Intrusion Detection System using CNN and LSTM deep learning architectures.

Each day was structured in three parts: a conceptual briefing by the mentor, a hands-on implementation session, and a documentation and review activity. This structure ensured that theoretical understanding was always immediately reinforced through practical application.

3.2 Phase 1: Cybersecurity and Networking Fundamentals (Days 1–16)

Days 1 through 7 covered cybersecurity theory and networking fundamentals: the CIA Triad, vulnerability assessment, OWASP Top 10, TCP/IP and OSI model security implications, IP addressing, DNS architecture, and HTTP/S with TLS. These sessions built the conceptual foundation for all subsequent practical work.

Days 8 through 16 transitioned to hands-on skill development. DNS enumeration using Nslookup, Dig, and Traceroute provided practical reconnaissance experience. Python scripting sessions produced functional tools for file hashing, integrity verification, log monitoring with Regex, and SSH automation. The Wireshark laboratory on Day 15 delivered direct experience with network traffic capture and payload analysis, while the web security assessment on Day 16 consolidated OWASP theory into authorized practical testing.

3.3 Phase 2: Keylogger Awareness Project (Days 17–23)

The second phase followed a structured software development lifecycle across seven days: requirements analysis, environment setup, core module development, Regex detection engine implementation, SQLite database integration, dashboard creation, and comprehensive testing with final documentation. Every activity was conducted within a strictly sandboxed environment, and the tool was designed exclusively for security awareness and educational demonstration.

3.4 Phase 3: AI-Based Intrusion Detection System (Days 24–26)

The third phase compressed the development of a fully functional AI-driven IDS into three intensive days. Day 24 covered IDS theory and exploratory data analysis of the benchmark datasets. Day 25 involved feature engineering, model implementation, and SMOTE-based class balancing. Day 26 focused on model evaluation using ROC-AUC, Confusion Matrix, precision, recall, and F1-score, as well as SHAP and LIME explainability analysis. Results were compiled into a comprehensive technical performance report.

3.5 Daily Activity Log

The table below provides a detailed account of all 26 days of the structured internship programme:

Day	Task Title	Key Activity & Analysis
1	Cybersecurity Fundamentals	Analyzed the CIA Triad and its application in risk evaluation and security policy design.
2	Vulnerability Assessment	Explored attack surfaces, threat vectors, CVSS scoring, and structured mitigation strategies.
3	OWASP Top 10	Studied the ten most critical web vulnerabilities with hands-on analysis of SQL Injection and XSS.
4	Networking Fundamentals	Evaluated TCP/IP and OSI model layers with focus on security implications at each layer.
5	IP Addressing & Subnetting	Practiced network segmentation using CIDR notation; explored how subnetting reduces attack surface.
6	DNS Concepts	Analyzed DNS architecture, resolver roles, and DNS-based attack vectors including DNS spoofing.
7	HTTP/S & TLS Security	Explored end-to-end encryption, X.509 certificates, and the TLS 1.3 handshake mechanism.
8	DNS Enumeration	Performed active reconnaissance using Nslookup, Dig, and Traceroute; documented findings.

9	Python Programming Refresher	Revised Python fundamentals: file I/O, exception handling, and data structures for security scripting.
10	Security Automation	Developed Python scripts for SHA-256 file hashing and automated file integrity verification.
11	Log Monitoring & Regex	Used Regular Expressions for parsing system logs, extracting events, and detecting suspicious patterns.
12	Advanced Security Tools	Created Python scripts for banner grabbing and service version auditing to identify vulnerable services.
13	Traffic Analysis Concepts	Explored anomaly detection techniques, baseline profiling, and network monitoring workflows.
14	SSH Automation	Developed Python scripts for automated SSH authentication and remote server administration.
15	Wireshark Laboratory	Captured and analyzed live network traffic using Wireshark to isolate anomalous TCP/UDP communication.
16	Web Security Assessment	Conducted authorized web application security testing in a sandboxed target and documented remediations.
17	Keylogger Project: Requirements	Defined project objectives, ethical boundaries, and technical requirements for the awareness tool.
18	Keylogger Project: Setup	Configured the Python development environment and implemented the keyboard event listener module.
19	Keylogger Project: Monitoring	Developed the core keystroke capture module capable of real-time event processing.
20	Keylogger Project: Regex Engine	Implemented Regex patterns for detecting passwords, email addresses, and OTPs in keystroke streams.
21	Keylogger Project: Database	Integrated SQLite for structured storage and retrieval of captured security events.
22	Keylogger Project: Dashboard	Created a visual analytics dashboard with alerts highlighting detected sensitive data exposure.
23	Keylogger Project: Testing	Validated detection accuracy across multiple test scenarios and finalized project documentation.
24	IDS Fundamentals	Studied signature-based vs. anomaly-based detection; reviewed NSL-KDD and UNSW-NB15 datasets.
25	IDS Model Development	Performed feature engineering; implemented and trained CNN and LSTM models with SMOTE balancing.
26	IDS Evaluation & Analytics	Tested model against simulated attacks; visualized ROC-AUC, Confusion Matrix, precision and recall.

3.6 Tools and Technologies Mastered

- Network Reconnaissance: Automated DNS reconnaissance using Python wrappers for Nslookup, Dig, and Traceroute, enabling efficient infrastructure mapping and attack surface identification.

- **Packet Analysis:** Deep analysis of TCP/UDP streams using Wireshark, including protocol anomaly identification, malicious payload isolation, and indicator of compromise (IoC) extraction.
- **Python Security Scripting:** File hashing, integrity checking, SSH automation, log parsing with Regex, and banner grabbing for service vulnerability identification.
- **Regex Pattern Matching:** Design and validation of multi-pattern Regex engines for real-time sensitive data detection including passwords, OTPs, email addresses, and financial identifiers.
- **Deep Learning for Security:** End-to-end ML pipeline from data preprocessing and feature engineering through CNN/LSTM model training to performance evaluation with explainability analysis.

CHAPTER 4 – PROJECT WORK / TECHNICAL LEARNING

4.1 Overview

The two major projects developed during this internship represent the practical culmination of all technical skills and conceptual knowledge built across the 26-day programme. The Keylogger Awareness Project addressed endpoint-level threat simulation and detection, while the Intrusion Detection System demonstrated how deep learning can detect and classify network-level intrusions. Together they provided a comprehensive, end-to-end perspective on both offensive threat simulation and AI-driven defensive response.

4.2 Project 1: Keylogger Awareness Project (Days 17–23)

The Keylogger Awareness Project was developed to demonstrate, within a fully controlled and ethically bounded sandboxed environment, the mechanisms by which malicious keylogging software operates, the types of sensitive data it can capture, and the defensive measures that mitigate its impact. All development and testing was conducted with explicit ethical constraints: the tool operated only on synthetic test inputs within an isolated environment and was never used to capture real personal data.

4.2.1 System Architecture

The project was designed as a modular, four-component pipeline: the Keystroke Monitoring Module captured raw keyboard events; the Sensitive Data Detection Engine applied Regex patterns to identify sensitive strings; the SQLite Database Storage Layer persisted events and alerts; and the Visualization Dashboard presented findings and triggered alerts. Each component was developed independently and then integrated into a cohesive real-time processing pipeline.

4.2.2 Keystroke Monitoring Module

Using Python's keyboard event handling library, an asynchronous event listener was implemented to capture all keyboard interrupts within the sandboxed environment. Each event was tagged with a timestamp, application context, and raw key value. A dedicated mapping module translated special keys (Backspace, Enter, Tab, function keys) into meaningful textual representations, ensuring the event log was accurate and human-readable for analysis and demonstration purposes.

4.2.3 Sensitive Data Detection Engine

The detection engine employed a library of carefully crafted Regular Expression patterns to identify the following sensitive data categories in real time:

- Passwords: Pattern `(?=.*[A-Z])(?=.*[a-z])(?=.*\d){8,}` — identifies strings with minimum 8 characters containing uppercase, lowercase, and digit characters.
- Email Addresses: Pattern `[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}` — detects email address entry, a common phishing and credential harvesting vector.
- One-Time Passwords (OTPs): Pattern `\b\d{6}\b` — identifies six-digit numeric sequences matching standard two-factor authentication OTP formats.
- Credit Card Numbers: Luhn algorithm-complemented Regex to detect potential credit card number entry sequences.
- Indian Mobile Numbers: Pattern `(?:\+91|0)?[6-9]\d{9}` — detects mobile number entry relevant to financial application security scenarios.

The engine was optimized to minimize false positives through contextual windowing — analyzing sequences of keystrokes rather than isolated characters — and confidence thresholds requiring multiple pattern matches within a defined time window before generating an alert.

4.2.4 Database Integration and Dashboard

An SQLite database with three tables — Events, Alerts, and Sessions — provided persistent structured storage for all captured data. The Visualization Dashboard consumed this data to generate real-time event counts, a timeline of detection alerts categorized by sensitive data type, a frequency heatmap of keystroke activity, and a color-coded alert feed distinguishing high-confidence from low-confidence detections.

The completed project demonstrated conclusively that unencrypted user inputs are highly vulnerable to capture and pattern-based analysis. Even a relatively simple Regex-based detection engine, applied to a continuous keystroke stream, can identify credentials, financial data, and authentication tokens with high reliability — reinforcing the critical importance of endpoint encryption, EDR solutions, and user awareness training.

4.3 Project 2: Machine Learning-Based Intrusion Detection System (Days 24–26)

The IDS project required integration of knowledge from multiple domains: network security fundamentals, statistical data analysis, machine learning engineering, deep learning architecture design, and AI model evaluation. The project demonstrated the transition from traditional rule-based detection to autonomous anomaly detection using two complementary deep learning architectures.

4.3.1 Dataset Analysis and Preprocessing

The NSL-KDD dataset contains 125,973 training and 22,544 test instances described by 41 features, classified into Normal traffic and four attack categories (DoS, Probe, R2L, U2R). The UNSW-NB15 dataset, generated using the IXIA PerfectStorm tool, contains approximately 2.5 million records with 49 features across nine attack categories, providing a more contemporary benchmark.

Preprocessing involved: handling missing values through median imputation; encoding categorical features (protocol type, service, flag) using one-hot encoding; normalizing continuous features using Min-Max scaling to [0,1]; and applying SMOTE to the training data to address the severe class imbalance between Normal traffic and rare attack categories such as U2R and R2L — precisely the attacks posing the greatest risk.

4.3.2 Model Architectures

Convolutional Neural Network (CNN)

The CNN model treated each network flow's feature vector as a one-dimensional spatial signal, enabling convolutional filters to detect local feature patterns indicative of specific attack types. Architecture: Input layer → two 1D Conv layers (64 and 128 filters, ReLU activation, Max Pooling) → Dropout (0.3) → Flatten → Dense layers (256, 128 neurons) → Softmax output for 5-class classification.

Long Short-Term Memory (LSTM)

The LSTM model captured temporal dependencies in sequential network traffic, enabling detection of attacks unfolding across multiple consecutive packets — such as slow-rate DoS and reconnaissance scans. Architecture: Input layer (sequences of flow records) → two stacked LSTM layers (128 and 64 units, return_sequences=True on first layer) → Dropout (0.25) → Dense classification head matching the CNN output structure.

4.3.3 Training, Evaluation, and Explainability

Both models were trained using the Adam optimizer (learning rate 0.001), categorical cross-entropy loss, batch size 256, over 30 epochs with learning rate reduction on plateau and early stopping callbacks. The 80/20 train-validation split ensured unbiased performance estimation.

SHAP values revealed that features related to network connection duration, byte transfer statistics, and protocol flags were most influential in distinguishing attack traffic. LIME generated locally faithful decision boundary approximations for individual flagged instances, enabling security analysts to understand why specific connections were classified as malicious — a critical operational requirement for real-world IDS deployment.

Performance was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC for each attack class versus the rest. The combination of SMOTE and LSTM architecture produced notably improved recall for minority attack classes (U2R, R2L), the categories most commonly missed by models trained without class balancing.

4.4 Summary of Technical Skills Acquired

Domain	Key Skills & Concepts Acquired
Cybersecurity Foundations	CIA Triad, OWASP Top 10, vulnerability assessment, TLS/SSL, DNS attack vectors, banner grabbing
Network Analysis	TCP/IP, OSI model, CIDR subnetting, DNS enumeration, Wireshark packet capture and payload analysis
Python Security Scripting	SHA-256 hashing, file integrity, SSH automation, log parsing, Regex pattern design
Keylogger Project	Async keystroke monitoring, multi-pattern Regex detection, SQLite schema, real-time dashboard
Machine Learning – IDS	Feature engineering, Min-Max scaling, SMOTE, CNN, LSTM, hyperparameter tuning
Model Evaluation	Confusion Matrix, ROC-AUC, Precision, Recall, F1-Score, SHAP and LIME explainability
Professional Skills	MITRE ATT&CK mapping, technical documentation, ethical hacking guidelines, report writing

CHAPTER 5 – FINDINGS AND LEARNING OUTCOMES

5.1 Key Technical Findings

Finding 1: The CIA Triad as a Universal Security Design Principle

Across every technical domain covered — from network configuration to ML model design — the principles of Confidentiality, Integrity, and Availability emerged as the consistent and universal evaluative framework. Whether assessing TLS implementations, designing the SQLite schema for the Keylogger project, or evaluating the IDS model's false negative rate (which directly impacts availability of security alerts), the CIA Triad provided a stable, practical analytical lens applicable throughout the engineering process.

Finding 2: Python as the Lingua Franca of Security Automation

Python's versatility and extensive library ecosystem make it uniquely suited for security automation. During the internship, a single language was used to accomplish an extraordinarily diverse range of security tasks: file integrity verification, SSH automation, DNS reconnaissance, log parsing with Regex, keylogger development, SQLite management, deep learning model development, and explainability analysis. No other language in the current landscape offers comparable breadth across security automation and AI simultaneously.

Finding 3: Class Imbalance is Critical in Security ML

Applying SMOTE to the NSL-KDD and UNSW-NB15 datasets revealed that class imbalance is not merely a statistical inconvenience but a fundamental challenge with serious practical implications. Without SMOTE, models exhibited severe recall deficits for minority attack classes (U2R, R2L) — precisely the attacks posing the greatest risk. This finding highlights the necessity of domain-specific evaluation metrics and imbalance handling strategies beyond standard accuracy optimization in security-focused ML systems.

Finding 4: Explainability is Operationally Essential for Security AI

The application of SHAP and LIME to the IDS model revealed a critical operational insight: in cybersecurity contexts, a high-accuracy model that cannot explain its predictions is operationally limited. Security analysts cannot act on an unexplained alert — they need to understand why a connection was flagged as malicious to determine the appropriate

response, escalate appropriately, and avoid alert fatigue. Explainable AI is therefore an operational necessity, not a research curiosity, for AI-driven security tools in real-world deployment.

Finding 5: Unencrypted Endpoint Inputs Remain a Critical Attack Vector

The Keylogger Awareness Project demonstrated with striking clarity that unencrypted user inputs remain a significant and often underestimated vulnerability in organizational security postures. Even a simple Regex-based detection engine applied to a continuous keystroke stream reliably identified passwords, OTPs, email addresses, and financial data — reinforcing the critical importance of endpoint encryption, application-level input sanitization, and full-disk encryption as essential components of a defense-in-depth architecture.

5.2 Skills Developed

Technical Skills

- Design and implementation of CNN and LSTM deep learning models for multi-class network traffic classification and intrusion detection.
- End-to-end ML data preprocessing pipelines: missing value imputation, feature encoding, Min-Max normalization, and SMOTE-based class balancing.
- Development of multi-pattern Regex sensitive data detection engines for real-time keystroke stream analysis.
- Network packet capture and traffic analysis using Wireshark, including TCP/UDP stream reconstruction and payload inspection.
- Python security automation: file integrity checking, SSH automation, DNS reconnaissance, log parsing, and banner grabbing.
- Application of SHAP and LIME explainability frameworks to deep learning security models for analyst-interpretable outputs.
- SQLite database design, schema creation, and query optimization for security event storage and correlation.

Professional Skills

- Technical report writing and documentation following academic and industry standards.
- Application of the MITRE ATT&CK framework to map simulated and observed adversary behaviors to documented tactics and techniques.

- Strict adherence to ethical guidelines in security research and tool development within sandboxed environments.
- Structured problem-solving following a software development lifecycle: requirements, design, implementation, testing, and documentation.

5.3 Industry Exposure and Professional Development

Beyond specific technical skills, the internship provided valuable exposure to professional cybersecurity norms and workflows. Working within a structured environment — with daily briefings, hands-on sessions, progress reviews, and documented deliverables — simulated the operational rhythm of a professional Security Operations Center (SOC) or AI security team.

Exposure to industry-standard toolsets such as Splunk Community Edition highlighted the significant gap between academic familiarity and professional proficiency. Working with real benchmark datasets provided a much more authentic understanding of real-world ML challenges — including data quality issues, class imbalance, feature relevance, and generalization across different network environments — than simplified academic toy datasets could provide.

The experience also underscored the critical importance of ethical conduct in cybersecurity practice. Every tool developed, every test conducted, and every dataset analyzed was approached with explicit awareness of ethical implications — a discipline that is as essential as any technical skill in the security profession. Responsible disclosure, minimal footprint testing, and sandbox isolation were treated not as optional formalities but as non-negotiable professional standards.

5.4 Knowledge Domains Consolidated

A comprehensive and interconnected understanding of the following knowledge domains was established and consolidated during the internship:

- Information Security Principles: CIA Triad, defense-in-depth strategy, principle of least privilege, threat modeling, and structured risk assessment frameworks.
- Network Security: TCP/IP and OSI security implications by layer, DNS architecture and spoofing attack vectors, TLS 1.3 handshake mechanics, CIDR-based network segmentation.

- Web Application Security: OWASP Top 10 vulnerabilities — SQL Injection, XSS, CSRF, insecure deserialization — and their mitigations in real application contexts.
- Machine Learning for Security: Supervised classification pipelines, deep learning architecture design (CNN, LSTM), class imbalance handling with SMOTE, and model evaluation with domain-appropriate metrics.
- Security Automation: Python scripting for defensive tasks, Regex-based event detection, log analysis, and SIEM fundamentals with Splunk Community Edition.
- Adversary Behavior Mapping: MITRE ATT&CK framework structure, tactic and technique identification, and mapping to specific defensive countermeasures.

5.5 Challenges Encountered and Solutions Applied

- Regex Precision vs. Recall Trade-off: Designing patterns precise enough to minimize false positives without sacrificing sensitivity required extensive iterative testing. The solution was confidence scoring and contextual windowing, analyzing keystroke sequences rather than isolated characters.
- LSTM Sequence Structuring: Restructuring tabular NSL-KDD feature data into sequential LSTM input format required a custom sliding window generator — a non-trivial data engineering challenge that deepened understanding of sequential model input requirements.
- Class Imbalance in Training Data: Extreme rarity of U2R and R2L attack samples caused initial models to exhibit near-zero recall for these critical classes. SMOTE resolved this, though careful validation was required to prevent data leakage between synthetic training samples and the test set.
- Ethical Constraints on Keylogger Testing: Ensuring all testing remained within the sandboxed environment required an input sanitization layer replacing any real personal text with synthetic test data during evaluation sessions.

CHAPTER 6 – CONCLUSION

6.1 Summary of the Internship

The 26-day summer internship at Logiccraft Software Solutions was an exceptional and transformative learning experience that successfully bridged the gap between the theoretical foundations of a BSc Computer Science education and the demanding practical realities of the professional cybersecurity industry. By progressing systematically from networking fundamentals and the CIA Triad through Python security automation and Regular Expression engineering to the implementation of advanced deep learning architectures for network intrusion detection, the programme provided a holistic and coherent education at the convergence of Artificial Intelligence and Cybersecurity.

The structured, three-phase curriculum demonstrated admirable pedagogical design: foundational knowledge was carefully established before being applied, and each new technical concept was immediately reinforced through hands-on implementation. The two major projects — the Keylogger Awareness System and the Machine Learning-based Intrusion Detection System — served as integrative capstones requiring the synthesis of skills from multiple domains, mirroring the complexity of real-world security engineering challenges.

6.2 Key Takeaways and Reflections

The most significant and enduring insight from this internship is the profound interconnection of all the domains it covered. Network security knowledge directly informed the IDS problem formulation; Python scripting skills enabled both the Keylogger project and the ML pipeline; Regex expertise powered the detection engine; database knowledge supported event persistence; and machine learning theory was grounded by direct engagement with benchmark datasets representing real network environments. Effective cybersecurity requires the ability to reason across all these layers simultaneously — from physical network to application layer, from raw data to model predictions, and from technical indicators to human-interpretable insights.

A particularly valuable insight was the understanding that security is never a binary state — it exists on a spectrum, determined by the cumulative effect of many design decisions, configurations, and operational practices. The MITRE ATT&CK framework made this

vivid: every technique in an attacker's playbook has a corresponding defensive countermeasure, and the strength of a security posture is measured by how comprehensively those countermeasures have been implemented, monitored, and maintained. This systems-level perspective on security is one that classroom instruction alone rarely conveys with the same clarity as direct practical experience.

On a personal level, the internship significantly deepened my appreciation for the ethical dimensions of cybersecurity work. Developing a keylogger — even for strictly educational purposes — required constant vigilance and an explicit ethical commitment. This reinforced the understanding that technical capability in cybersecurity must always be coupled with a strong ethical framework, and that the discipline's most instructive tools are also its most dangerous when approached without appropriate responsibility.

6.3 Future Directions

- Pursuing industry certifications such as CompTIA Security+, Certified Ethical Hacker (CEH), or EC-Council's Certified Network Defender (CND) to formally validate the practical skills developed during the internship.
- Deepening expertise in AI-driven threat hunting by exploring advanced topics including Graph Neural Networks (GNNs) for network traffic analysis and Transformer architectures for security log analysis.
- Gaining proficiency in cloud security — particularly AWS and Azure security configurations, IAM best practices, and cloud-native SIEM solutions.
- Participating in Capture the Flag (CTF) competitions and contributing to open-source security tools to continue developing practical adversarial thinking and defensive engineering skills.
- Exploring the rapidly evolving field of Adversarial Machine Learning — understanding how AI security models can be attacked through data poisoning and adversarial examples, and how such attacks can be defended against.

6.4 Final Observations

The summer internship at Logiccraft Software Solutions has been a defining experience in my academic journey. It has transformed my understanding of cybersecurity from an abstract collection of principles and tools into a coherent, living discipline with urgent real-world relevance. The skills, knowledge, and professional perspectives gained during these

26 days have provided both the technical foundation and the intellectual confidence to pursue a meaningful career at the intersection of Artificial Intelligence and Cybersecurity.

I am deeply grateful to Logiccraft Software Solutions, to my Faculty Guide Ms. Gayathri A., and to the entire faculty and administration of Naipunnya School of Management for making this experience possible. The investment of time, expertise, and guidance by all involved has laid a foundation that I am committed to building upon throughout my professional career.

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APPENDICES

Appendix A: Internship Completion Certificate from Logiccraft Software Solutions.

Appendix B: Daily Attendance Sheet and Activity Log.

Appendix C: Python Source Code – Sensitive Data Detection Engine (Regex Patterns and Keystroke Monitor).

Appendix D: Python Source Code – IDS Model Training Pipeline (Feature Engineering, CNN/LSTM Implementation, SMOTE).

Appendix E: Visualization of IDS Performance Metrics (Confusion Matrix, ROC-AUC Curves, SHAP Feature Importance Plots).

Appendix F: NSL-KDD and UNSW-NB15 Dataset Feature Descriptions and Class Distribution Statistics.